

Ratio & Proportion – APTITUDE LEARNING NOTES

Complete concepts • formulas • shortcuts • solved examples • practice

1. Ratio Basics

Ratio compares quantities in the same units. $a:b = a/b$.

If $a:b=m:n$, write $a=mk$ and $b=nk$.

FORMULA: $a:b=c:d \Leftrightarrow ad=bc$.

2. Division in a Given Ratio

If total T is divided in ratio $m:n$, there are $m+n$ equal parts.

FORMULA: First part $= T \times m / (m+n)$; second part $= T \times n / (m+n)$.

3. Compound Ratio

Compound ratio of $a:b$ and $c:d$ is $ac:bd$.

Always simplify the final ratio when possible.

4. Proportion

$a:b=c:d$ means the two ratios are equal.

FORMULA: Product of extremes = product of means $\rightarrow ad=bc$.

5. Direct Proportion

If $x \propto y$, x/y is constant.

FORMULA: $x_1/y_1 = x_2/y_2$.

6. Inverse Proportion

If $x \propto 1/y$, xy is constant.

FORMULA: $x_1 y_1 = x_2 y_2$.

7. Partnership

Profit share is proportional to capital $\times$ time.

FORMULA: Share ratio $= C_1 T_1 : C_2 T_2$.

8. Mixture / Allegation

If values a and b are mixed to obtain mean m , quantities are in ratio $(b-m):(m-a)$, for $a < m < b$.

This is useful for concentration, price, and mixture problems.

9. Mean and Third Proportional

FORMULA: Mean proportional between $a, b = \sqrt{ab}$.

FORMULA: Third proportional to $a, b = b^2/a$.

Solved Aptitude Examples

Q1. Divide ■840 in ratio 3:4.

Solution: 7 parts → ■360 and ■480.

Q2. If $x:y=5:7$ and $y:z=14:9$, find $x:y:z$.

Solution: Scale 5:7 to 10:14 → **10:14:9**.

Q3. 8 workers complete a job in 15 days. How many days for 12 workers?

Solution: $8 \times 15 = 12d \rightarrow$ **10 days**.

Q4. 6 pens cost ■90. Cost of 14?

Solution: $90 \times 14 / 6 =$ **■210**.

Q5. Partners invest ■4000×12 months and ■6000×9 months. Divide ■700.

Solution: $48000:54000=8:9 \rightarrow$ **■329.41 and ■370.59**.

Q6. Boys:girls=3:2, total 45.

Solution: **27 boys, 18 girls**.

Q7. Fourth proportional to 4,6,8.

Solution: $4:6=8:x \rightarrow$ **12**.

Q8. Mean proportional between 9 and 25.

Solution: $\sqrt{225}=15$.

Q9. 20 L mixture milk:water=3:2.

Solution: Milk=**12 L**, water=**8 L**.

Q10. 30% and 50% solutions give 40%. Ratio?

Solution: $(50-40):(40-30)=1:1$.

Practice Questions

1. Divide ■960 in ratio 5:7.

2. If $a:b=3:4$ and $b:c=8:9$, find $a:b:c$.

3. 10 workers do a job in 18 days. Find days for 15 workers.

4. 7 notebooks cost ■210. Find cost of 12.

5. A and B invest ■6000 for 10 months and ■9000 for 8 months. Divide ■720 profit.

6. Ages A:B=4:5; total=54. Find ages.

7. Find fourth proportional to 5,10,20.

8. Find mean proportional between 16 and 36.

9. A 24 L mixture has acid:water=5:3. Find each.

10. 20% and 50% solutions are mixed to obtain 30%. Find ratio.

11. If $3:5=x:45$, find x .

12. If 4 men or 6 women can do a job in 12 days, compare one man's and one woman's efficiency.

Answer Key

1) ■400, ■560 • 2) 6:8:9 • 3) 12 days • 4) ■360 • 5) ■360, ■360 • 6) 24, 30 • 7) 40 • 8) 24 • 9) 15 L, 9 L • 10) 2:1 • 11) 27 • 12) Man:woman=3:2

Quick Revision Formula Sheet

- $a:b=c:d \Leftrightarrow ad=bc$
- T in $m:n \rightarrow Tm/(m+n), Tn/(m+n)$
- Direct: $x■/y■=x■/y■$
- Inverse: $x■y■=x■y■$
- Partnership: $\text{capital} \times \text{time}$
- Mean proportional $= \sqrt{ab}$
- Allegation $= (\text{higher} - \text{mean}) : (\text{mean} - \text{lower})$

Exam strategy: Identify the pattern first, choose the shortest valid formula, solve step-by-step, and verify the final answer.